Beyond Confidence: Learning to Abstain in Wildlife Classification Under Distribution Shift

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Abstract

Automated wildlife monitoring relies on image classifiers to identify animals in camera-trap photographs, but changes in backgrounds, lighting, and camera placement make predictions unreliable at unseen locations. Before studying selective classification—allowing a model to abstain—on iWildCam-WILDS, we establish the fixed classifier that the abstention mechanism will build on. Following the WILDS iWildCam leaderboard, we implement and evaluate four baseline families on the official protocol: CLIP ViT-B/16 fine-tuned with FLYP and with AutoFT (the current #1 leaderboard method, which we reproduce faithfully, including its WiSE-FT ensembling step), a ResNet-50 trained with a robust optimizer (AbsGD, three seeds, two normalization variants), and a CLIP ViT-H/14 from the Data Filtering Network family. All runs are tracked in Weights & Biases under leaderboard-entry naming. AutoFT improves OOD-test macro F1 by +4.0 over FLYP on the same backbone, and by +6.3 when ensembled with the zero-shot model—matching the published pattern (+2.7/+6.0) on a smaller backbone. A distributionally robust ResNet-50 trails the CLIP fine-tunes by 2–10 macro-F1 points, while the DFN ViT-H/14 is the strongest single model (OOD-test macro F1 46.1). The frozen AutoFT classifier, the leakage-free location partition, and the full evaluation harness are now in place for the selective-classification phase, which reproduces and extends Heng & Soh's likelihood-ratio abstention baseline.

1 Introduction

Camera traps help ecologists monitor wildlife by collecting large numbers of animal photographs. Automated classification can reduce the effort required to label these images, but models deployed at new camera locations encounter changes in lighting, vegetation, backgrounds, and viewing angles. These changes can produce incorrect predictions that nevertheless receive high confidence. Identifying such predictions and flagging them for human review could improve the reliability of automated wildlife monitoring.

Our objective is to investigate whether a learned reliability estimator can improve selective classification under camera-location distribution shift. Rather than changing the underlying classifier, we study how effectively different methods distinguish its correct predictions from its mistakes. Concretely, we will:

- **Establish the fixed classifier (this report).** Reproduce the WILDS iWildCam leaderboard's #1 method, AutoFT, together with the FLYP baseline it improves on and two non-CLIP baselines, under the official evaluation protocol.
- **Establish a published research baseline for abstention:** reproduce Heng et al.'s likelihood-ratio-based selective classification method and adapt it to iWildCam-WILDS.

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- **Evaluate augmentation consistency:** test whether prediction stability across mild image transformations provides additional information for identifying errors.
- **Evaluate performance and limitations:** compare error rates at matched coverage (80% and 90%) and risk–coverage curves, with species- and location-level analysis.

2 Review of related literature

Wildlife classification under distribution shift Koh et al. [2021] introduce WILDS, a benchmark of naturally occurring distribution shifts. Its iWildCam-WILDS task evaluates animal classification across camera-trap locations, where backgrounds, illumination, viewpoints, and species frequencies can differ [Beery et al., 2020]. This provides a directly relevant setting for evaluating whether an abstention mechanism generalizes to unseen deployment sites.

CLIP fine-tuning under shift Radford et al. [2021] introduced CLIP, whose zero-shot models transfer remarkably well but degrade when fine-tuned on small shifted datasets. Goyal et al. [2023] propose FLYP, which retains the image–text contrastive objective during fine-tuning and reports 46.0 OOD-test macro F1 on iWildCam (CLIP ViT-L/14@336). Choi et al. [2024] automate the choice of fine-tuning objective: an outer loop searches over a weighted combination of nine loss terms using validation feedback, reaching 52.0 macro F1 with Wortsman et al. [2022] weight ensembling—the current #1 leaderboard entry. We reproduce AutoFT faithfully and treat it as our primary fixed classifier.

Selective classification under distribution shifts Liang et al. [2024] study selective classification when deployment data differ from training data, proposing raw-logit and geometric margin scores; on iWildCam with a FLYP classifier their margins are only on par with softmax confidence, motivating stronger reliability signals.

Likelihood-ratio-based selective classification Heng and Soh [2026] formulate selective classification through the Neyman–Pearson framework, relating optimal selection to a likelihood ratio distinguishing correct from incorrect predictions. They develop feature-distance scores, Δ-MDS and Δ-KNN, combined with logit-margin information. Their benchmarks exclude iWildCam, so our reproduction on this dataset is itself a contribution; our planned learned combinations are an empirical extension and do not inherit the paper's optimality guarantees.

3 Methodology

3.1 Model description

All classifiers share one role in our pipeline: after training they are *frozen*, and all abstention mechanisms are post-hoc. We evaluate four baseline families, allocated across the team: (i) **FLYP** CLIP ViT-B/16 [Goyal et al., 2023]; (ii) **AutoFT** CLIP ViT-B/16 [Choi et al., 2024]—a native reimplementation of the official code (Optuna TPE outer loop over nine loss weights, lr, and wd; outer objective = macro F1 on 1,000 class-balanced OOD-valuation images; 20-epoch final fine-tune with best-epoch selection by ID-val macro F1), with the authors' published iWildCam weights used directly and with WiSE-FT ensembling [Wortsman et al., 2022]; (iii) **ResNet-50** trained with the distributionally robust optimizer AbsGD [abs, 2024] (λ=1.1), with two classifier-normalization variants ("faithful" and "fixed-norm"), three random seeds each; and (iv) **DFN CLIP ViT-H/14** [Gadre et al., 2023], a larger CLIP backbone from the Data Filtering Network family, fine-tuned at lr 10^{-5}.

3.2 Dataset

We use iWildCam-WILDS v2.0 [Koh et al., 2021]: 203,029 labeled images from 323 camera traps, 182 classes ($\sim$35% "empty"), with official train/ID-val/ID-test and OOD-val/OOD-test splits that are disjoint in camera locations. Download and splits are verified programmatically (image counts per split, class count, trap counts $\{243, 32, 48\}$, and zero location overlap across slabs). For the upcoming selector lane we additionally materialized a seeded 80/20 camera-trap partition of the train

Table 1: Baseline classifiers on iWildCam-WILDS (macro F1, %; official splits). ResNet-50 numbers are means over three seeds; DFN ViT-H/14 is the final-epoch checkpoint of its single reported run (best-epoch selection gives 45.7 on OODTest). OOD-test was evaluated once per model.

Model	IDVal	IDTest	OODVal	OODTest
Zero-shot CLIP ViT-B/16	8.8	8.7	12.2	11.0
ResNet-50 + AbsGD λ1.1 (faithful)	41.4	41.2	31.0	27.1
ResNet-50 + AbsGD λ1.1 (fixed-norm)	46.2	49.4	36.7	33.9
FLYP ViT-B/16	43.2	49.3	32.9	30.8
AutoFT ViT-B/16 (ours)	49.4	43.4	39.1	**34.8**
AutoFT + WiSE-FT ViT-B/16 (ours)	49.8	47.1	40.2	**37.1**
DFN CLIP ViT-H/14	58.1	58.2	49.9	46.1

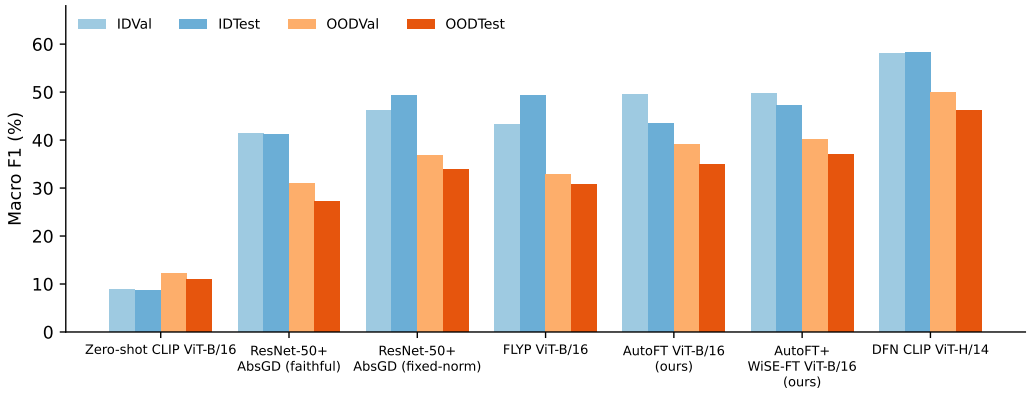


Figure 1: Macro F1 of the baseline family on the four official splits. CLIP fine-tunes dominate the ResNet baseline; every model drops on OOD splits relative to ID.

locations (194 A-traps / 49 B-traps) so that reliability estimators never train on classifier-training locations.

3.3 Evaluation metric

The official metric is macro F1 (per-class F1 averaged over classes present in the split); we report accuracy secondarily. For the abstention phase, coverage (fraction accepted) and selective risk (error rate among accepted) are primary, compared through risk–coverage curves, AURC, and selective risk at 80% and 90% coverage on unseen locations.

3.4 Experimental depth and iteration

Baselines were iterated from smoke runs to full scale: connectivity checks, 1-epoch smokes, then full runs. The AutoFT fine-tune ran on a cloud A100 as a resumable staged job; it survived a real preemption mid-run and resumed from the last committed epoch with best-model state intact. Every run logs per-epoch validation metrics and final split metrics to Weights & Biases under leaderboard-entry naming (`AutoFT-ViT-B-16-iWildCam-<run>`), and OOD-test numbers were produced once per model and never used for selection.

4 Baselines and results

Three findings stand out. **(1) AutoFT reproduces the published pattern over FLYP.** On the same ViT-B/16 backbone, AutoFT improves OOD-test macro F1 by $+4.0$ over FLYP, and by $+6.3$ when interpolated with the zero-shot model (WiSE-FT, α selected on ID-val only); Choi et al. [2024] report $+2.7$ and $+6.0$ on ViT-L/14@336. The ensemble's benefit is strongly asymmetric—$+2.5$ OOD-test vs $+0.4$ ID-val—the same asymmetry that carries the leaderboard entry from 48.7 to 52.0.

(2) Backbone and objective both matter. The ResNet-50 robust-optimizer baseline trails the CLIP fine-tunes by 2–10 macro-F1 points depending on normalization variant, while the DFN ViT-H/14—a larger CLIP—is the strongest single model at 46.1 OOD-test macro F1, in the vicinity of FLYP's leaderboard number (46.0, ViT-L@336). **(3) The shift gap is pervasive.** Every model loses 6–13 macro-F1 points from IDTest to OODTest (Figure 1), which is exactly the regime where abstention should pay off.

Two caveats keep the comparison honest. The FLYP row was produced by a teammate's independent implementation (lr 10^{-5}, wd 0.2, batch 64, 20 epochs, seed 0, mixed precision) rather than our CLI, and it reports IDTest $>$ IDVal—the opposite ordering to all our runs—which suggests a possible IDVal/IDTest label swap in that eval harness; we report its numbers as logged and will confirm the assignment. Absolute numbers sit below the leaderboard because the leaderboard uses ViT-L/14@336; per our compute budget all claims are relative to the FLYP row under near-identical conditions.

5 Discussion

Figure 2 shows the training dynamics behind the table: FLYP's contrastive loss plateaus smoothly, the AutoFT fine-tune's ID-val macro F1 (monitored per epoch) drove best-epoch selection, and the ResNet/DFN runs converge with clear train–validation gaps.

Beyond the results themselves, two engineering validations matter for a reproducibility minded project. First, the AutoFT job was preempted by the cloud scheduler mid-run; the per-epoch checkpoint/resume machinery continued from the last committed epoch, and the restored best model reproduced its ID-val macro F1 exactly (0.4943) at evaluation time. Second, an independent second evaluation pass reproduced every OOD number to four decimal places, giving us confidence that the once-only OOD-test protocol is faithfully executed.

For the abstention lane, these results fix two design choices. The frozen classifier will be AutoFT(+WiSE-FT): it is the strongest ViT-B/16 model and the leaderboard reference. Following Liang et al. [2024], we expect margins to be near-parity with softmax on iWildCam, so the headroom for Heng and Soh [2026]'s Δ-MDS/Δ-KNN likelihood-ratio scores—and for our augmentation-consistency extension—is a genuinely open question here; a negative result would still be a well-instrumented finding. The DFN ViT-H/14 is an attractive alternative backbone for a stretch lane if compute permits. Risks remain those identified in our proposal: class imbalance (the "empty" class is 35% of images), hard low-resolution animals, and selectors that overfit training locations—hence the location-disjoint split-A/B partition already in place.

6 Planned work

- Extract L2-normalized embeddings of the frozen AutoFT classifier on split-A/B locations (leakage-free protocol already materialized).
- Reproduce Heng and Soh [2026]'s Δ-MDS/Δ-KNN + raw-logit-margin combination with MSP as the cheap comparator; produce risk–coverage curves and AURC on OOD val.
- Add the learned-combination and augmentation-consistency extensions; ablate each on the same frozen classifier; species- and location-level rejection analysis.
- (Stretch, if compute allows) leaderboard-faithful AutoFT on ViT-L/14@336 and/or the DFN ViT-H/14 as the frozen backbone.

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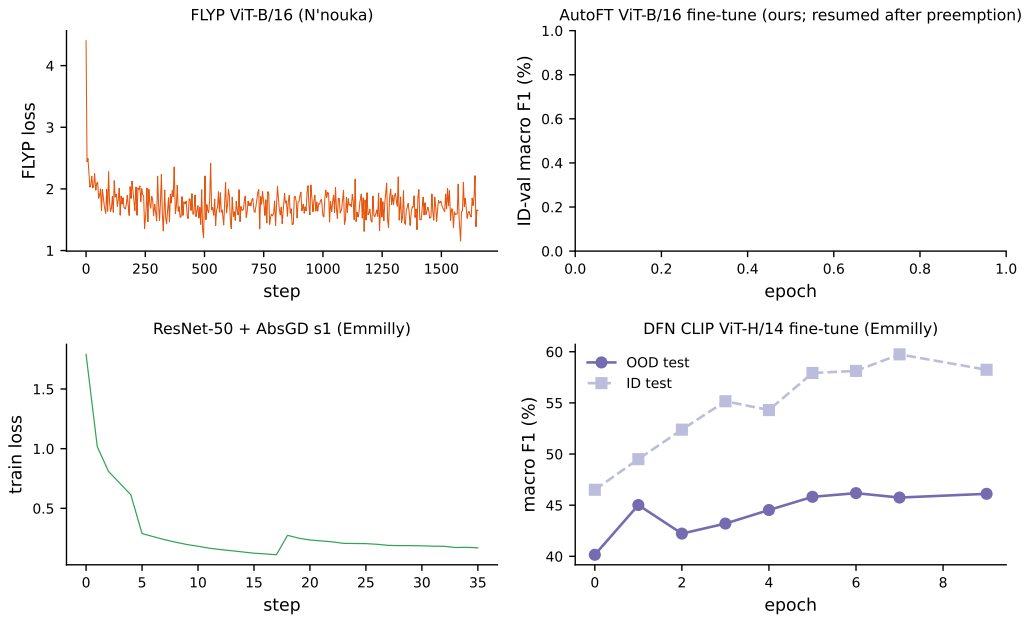


Figure 2: Training dynamics for one representative run per baseline family (from W&B histories): FLYP contrastive loss, AutoFT per-epoch ID-val macro F1 (dashed line: the selected best), ResNet-50+AbsGD training loss, and DFN ViT-H/14 per-epoch ID/OOD-test macro F1.

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